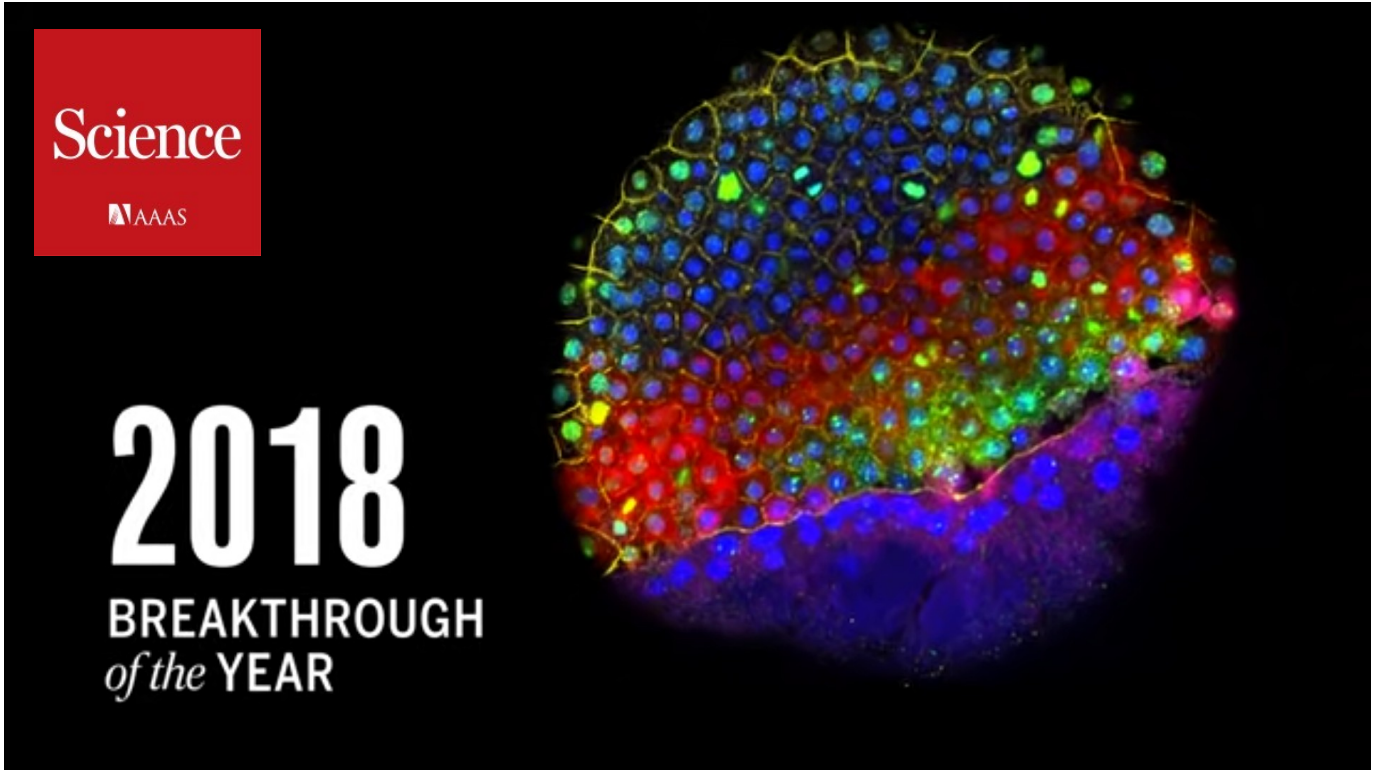
Single Cell RNAseq Analysis

Nikolay Oskolkov, Metabolic Research Group leader at NIRI, Riga, Latvia

PR Stats course, **Session 2: Pre-processing and quality control of scRNAseq data**



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Nature method of the year:
2019 (single cell multi-omics),
2020 (spatial transcriptomics)

Image adapted from Science, 362, 6421, 2018

Session content

Topics we'll cover in this session:

1) Filtering cells and genes: cell cycle and doublets

2) Selecting highly-variables genes

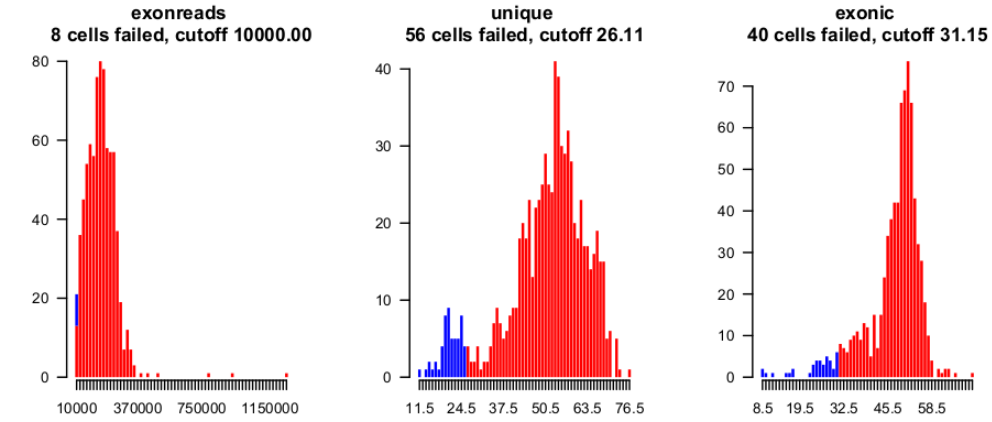
3) Practical demonstration of Seurat computational scRNAseq workflow

Quality Control:

1) Cell filtering
2) Gene filtering

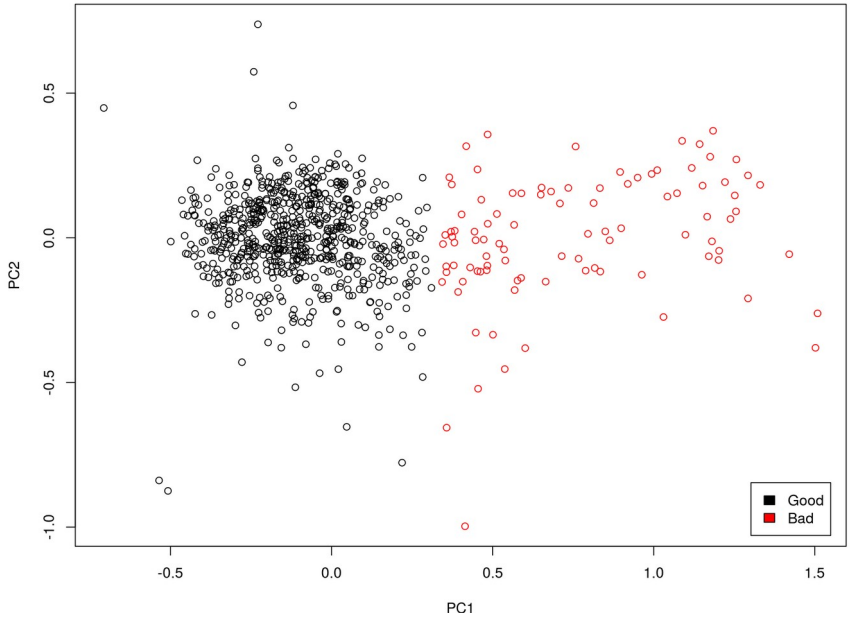
Cell Filtering

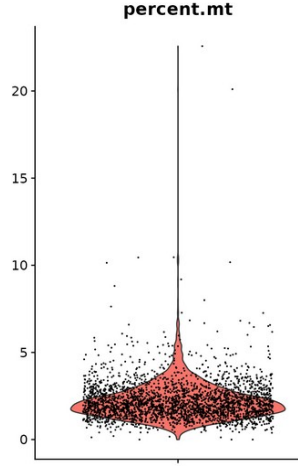
Univariate filtering: STAR stats



total 52 samples failed QC
Total samples: 768

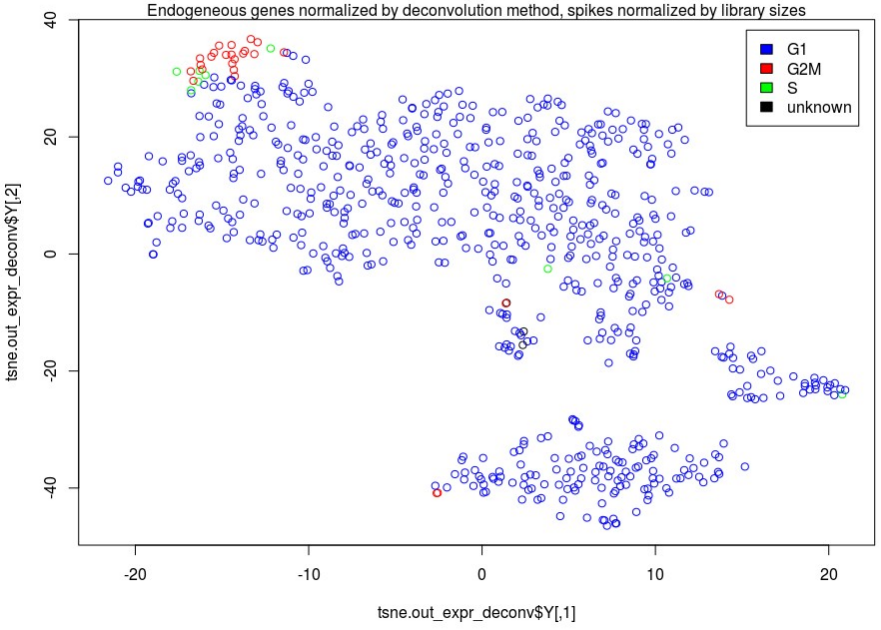
Multivariate filtering: SCATER





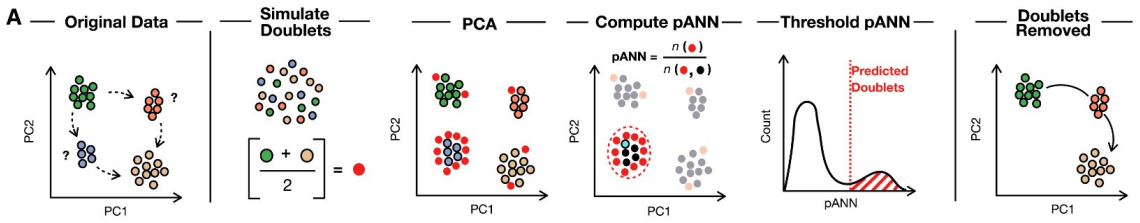
Removing cells with large mitochondrial RNA counts – these cells are low quality (dying or broken membrane cells)

Cell Filtering

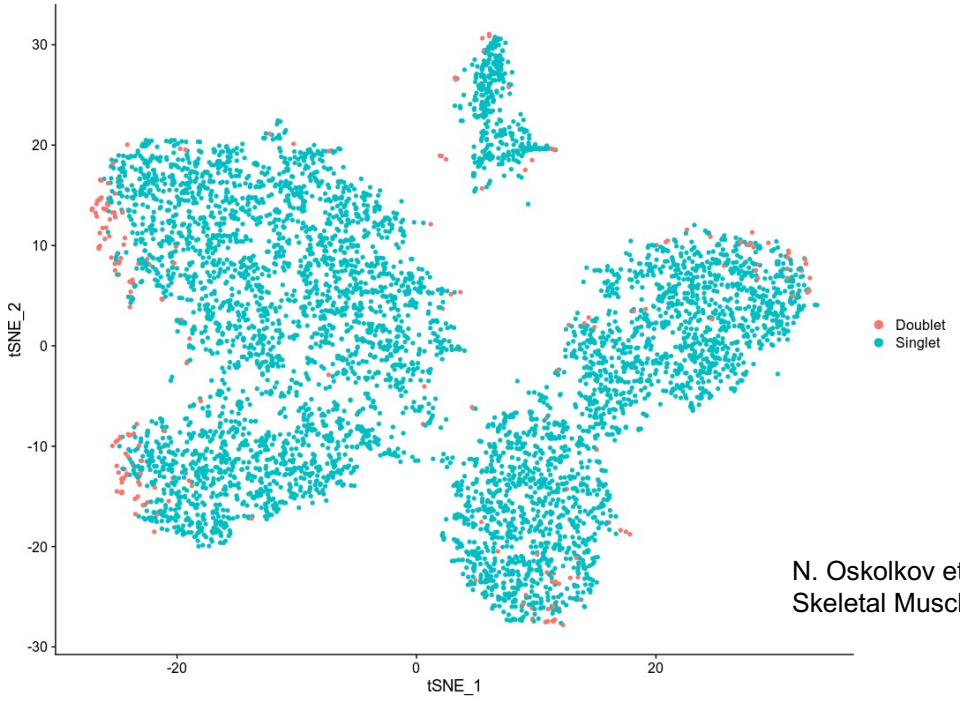


Keep an eye on cycling cells, they sometimes might form separate clusters: CYCLONE

Doublets sometimes are important to remove: DoubletFinder



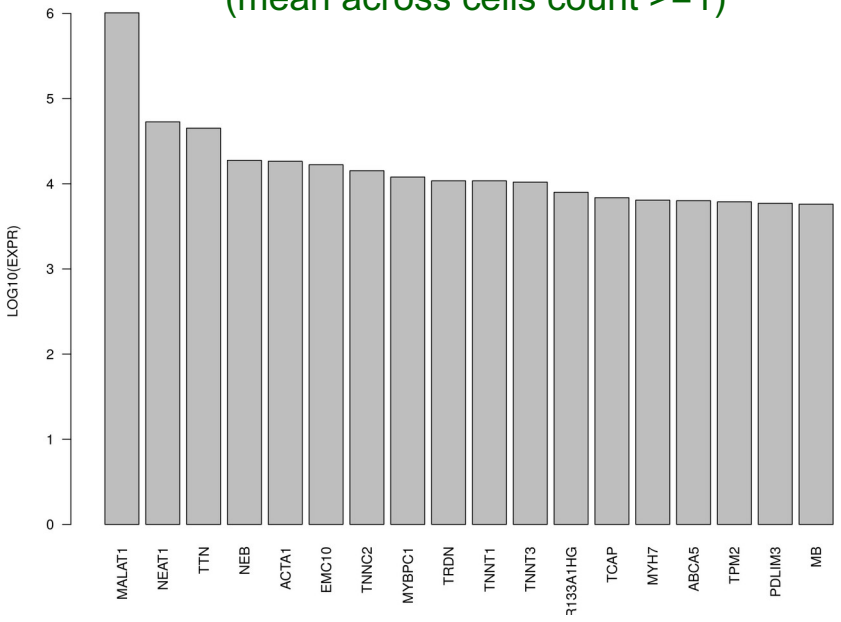
McGinnis et al., Cell Syst. 2019



N. Oskolkov et al., Skeletal Muscle 2022

Gene Filtering

Inherited from bulk RNAseq: remove lowly-expressed genes (mean across cells count >=1)



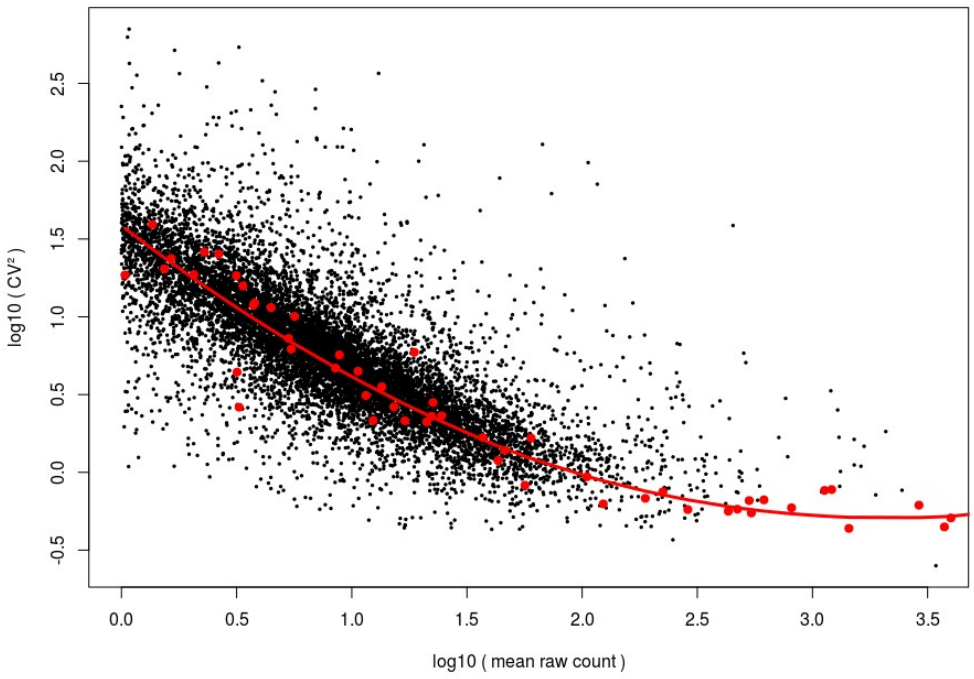
MALAT1 is a nuclear lincRNA, which typically indicates broken cell membrane and correlates with mitochondrial contamination

Typically MALAT1 is removed in order not to bias other genes counts

Removing ribosomal house keeping genes

```
expr<-expr[grepl(paste0(c("^RPS","^RPL"),collapse="|"),rownames(expr))==FALSE,]
```

Selecting highly-variable genes



Session summary

Take home messages of the session:

1) Filtering across cells is critical for successful identification of cell populations

2) Selecting highly-variables genes provides cleaner and more robust information about cell clusters

National Bioinformatics Infrastructure Sweden (NBIS)



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Vetenskapsrådet

Lunds Universitet

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2027 National Development Plan

Funded by the European Union

Acknowledgments: Metabolic Research Group



Open positions: fully covered PhD and 3-year postdoctoral fellow positions!